Job Leon Feldbrugge

National Taiwan University
Leung Center for Cosmology and Particle Astrophysics
Chee-Chun Leung Cosmology Hall
No. 1, Sec. 4, Roosevelt Road
Taipei 10617, Taiwan

jobfeldbrugge@ntu.edu.tw
jfeldbrugge.github.io

Referees

Prof. Neil Turok, University of Edinburgh, Edinburgh, United Kingdom and Perimeter Institute for Theoretical Physics, Waterloo, Canada (neil.turok@ed.ac.uk)
Prof. Ue-Li Pen, Academia Sinica, Taipei, Taiwan and Canadian Institute for Theoretical Astrophysics, Toronto, Canada (pen@cita.utoronto.ca)
Prof. Celine Boehm, University of Sydney, Sydney, Australia (celine.boehm@sydney.edu.au)
Prof. Rien van de Weygaert, University of Groningen, Groningen, The Netherlands (weygaert@astro.rug.nl)

Academic positions

2026 - present **Member and Postdoctoral Fellow** (joint position) at the Institute for Advanced Study, Princeton, United States of America and the Leung Center for Cosmology and Particle Astrophysics, National Taiwan University, Taipei, Taiwan

2021 - 2026 **Higgs Fellow** at the Higgs Centre for Theoretical Physics at the University of Edinburgh, Edinburgh, United Kingdom

2019 - 2021 **Postdoctoral Fellow** (joint position) at the Perimeter Institute, Waterloo, Canada and Carnegie Mellon University, Pittsburgh, United States of America

Areas of specialization

Applied mathematics: asymptotic analysis and resurgence, catastrophe theory, oscillatory integrals and Picard–Lefschetz theory • Scientific computing and machine learning • Wave optics and inverse problems • Lagrangian fluid dynamics and the cosmic web • Real-time path integrals and mathematical physics

Education

2015 - 2019 **PhD** Physics, Perimeter Institute, University of Waterloo
Advisor: Neil Turok
Thesis: Path integrals in the sky: Classical and quantum problems with minimal assumptions
Defense date: October 17, 2019

2014 - 2015 **Master** Part III Mathematics (*with distinction*), University of Cambridge
Essay supervisors: Paul Shellard and Tommaso Giannantonio
Essay: Primordial non-Gaussianity and large-scale structure

2012 - 2014 **Master** Physics (*cum laude*), Van Swinderen Institute, University of Groningen

2012 - 2014 **Master** Astronomy (*cum laude*), Kapteyn Institute, University of Groningen

2012 - 2014 **Master** Mathematics (*cum laude*), Bernoulli Institute, University of Groningen
Committee: Rien van de Weygaert (cosmology and large-scale structure formation)
Diederik Roest (string cosmology)
Aernout van Enter (statistical mechanics)
Thesis: Statistics of caustics in large-scale structure formation

2009 - 2012 **Bachelor** Physics (*cum laude*), Van Swinderen Institute, University of Groningen

2009 - 2012 **Bachelor** Astronomy (*cum laude*), Kapteyn Institute, University of Groningen

2009 - 2012 **Bachelor** Mathematics (*cum laude*), Bernoulli Institute, University of Groningen

Committee: Rien van de Weygaert (cosmology and large-scale structure formation)
Elisabetta Pallante (quantum field theory)
Gert Vegter (computational geometry)
Thesis: Analysis of Betti numbers and persistence diagrams of 2D Gaussian random fields

July 2016 **Summer school** It from qubit summer school, Perimeter Institute, Waterloo, Canada (two weeks)
July 2015 **Summer school** Prospects in theoretical physics: new insights into quantum matter, Institute for Advanced Study, Princeton (one week)
July 2015 **Summer school** Princeton summer school on condensed matter physics, Princeton University and Institute for Advanced Study, Princeton (one week)
August 2011 **Summer school** Dealing with environmental heritage, Bath, United Kingdom (two weeks)
July 2010 **Summer school** University of Cambridge international summer school in science (two weeks)

Awards

April 2020 Canadian Association of Physicists' Division of Theoretical Physics (DTP) and Winnipeg Institute for Theoretical Physics (WITP) P.R. Wallace PhD Thesis Prize
Best PhD thesis in Theoretical Physics from a Canadian university (2020).
November 2014 De Zeeuw-Van Dishoeck Award 2014
Best master thesis in Astronomy from a Dutch university (2014).
July 2014 GUF-100 prize 2014
Best student in the Faculty of Mathematics and Natural Sciences at the University of Groningen (2014).
November 2011 Silver medal in the University Physics Competition 2011
A university competition in which groups of three physics students all over the world solve a problem and write an article in 48 hours.
September 2010 Young Talent encouragement prize 2010 in Physics
Prize awarded by the Koninklijke Hollandse Maatschappij der Wetenschappen (Royal Holland Society of Sciences and Humanities) for the best freshman Physics student at the University of Groningen in 2009-2010.

Scholarships

March 2015 University of Waterloo scholarship, for PhD at the Perimeter Institute
July 2014 Hendrik Muller Fund 2014: Scholarship for excelling Dutch students
June 2014 VSB Fund 2014: Scholarship for Dutch students studying abroad

Research visits

November 2025 Research visit to Chris Howls and Inês Aniceto, University of Southampton, Southampton, United Kingdom
October 2025 Research visit to Sandrine Codis, Université Paris-Saclay, Paris, France
September 2025 Research visit to Chris Howls and Inês Aniceto, University of Southampton, Southampton, United Kingdom
May 2023 Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
March 2023 Research visit to Ue-Li Pen, Academia Sinica Institute of Astronomy and Astrophysics, Taipei, Taiwan
March 2023 Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
January 2023 Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
June 2022 Research visit to Rien van de Weygaert, Kapteyn Institute, Groningen, The Netherlands
January 2020 Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
June 2019 Research visit to Rien van de Weygaert, Kapteyn Institute, Groningen, The Netherlands
October 2018 Research visit to Rien van de Weygaert, Kapteyn Institute, Groningen, The Netherlands
May 2018 Research visit to Jean-Luc Lehners, Albert Einstein Institute, Potsdam, Germany
March 2017 Research visit to Jean-Luc Lehners, Albert Einstein Institute, Potsdam, Germany
Feb - Apr 2017 Research visit to DAMTP, University of Cambridge, United Kingdom

Publications & talks

Refereed journal articles and proceedings

2026 B. Hertzsch, **J. Feldbrugge**, and R. van de Weygaert, "Galaxy populations in the IllustrisTNG caustic skeleton," *Monthly Notices of the Royal Astronomical Society* 552, stag1724 (2026). arXiv:2604.18209 [astro-ph.CO]

2026 **J. Feldbrugge**, S. Crew, and U.-L. Pen, "Resurgence and hyperasymptotics in wave optics astronomy," *Journal of Cosmology and Astroparticle Physics* 2026(09), 022 (2026). arXiv:2602.21493 [astro-ph.CO]

2026 **J. Feldbrugge** and U.-L. Pen, "The real-time Feynman path integral for step potentials," *Annals of Physics* 490, 170490 (2026). arXiv:2508.17578 [quant-ph]

2026 A. Weber, **J. Feldbrugge**, and E. Pisanty, "Universal approach to saddle-point methods in attosecond science," *Physical Review A* 113, 063111 (2026). arXiv:2510.12545 [quant-ph]

2026 B. Hertzsch, **J. Feldbrugge**, M. Rodriguez, and R. van de Weygaert, "A new recipe for caustic pancakes: on the reality of walls in the cosmic web," *Journal of Cosmology and Astroparticle Physics* 2026(02), 037 (2026). arXiv:2510.02419 [astro-ph.CO]

2025 **J. Feldbrugge** and R. van de Weygaert, "What makes a cosmic filament? The dynamical origin and identity of filaments – I. Fundamentals in 2D," *Monthly Notices of the Royal Astronomical Society* 539, 873–901 (2025). arXiv:2405.20475 [astro-ph.CO]

2025 **J. Feldbrugge**, D. L. Jow, and U.-L. Pen, "Complex classical paths in quantum reflections and tunneling," *Physical Review D* 111, 085027 (2025). arXiv:2309.12420 [quant-ph]

2025 **J. Feldbrugge** and J. Y. L. Jones, "Efficient evaluation of real-time path integrals," *Physical Review D* 111, 083524 (2025). arXiv:2501.16323 [quant-ph]

2025 B. Bonga, **J. Feldbrugge**, and A. Metidieri, "Wave optics for rotating stars," *Physical Review D* 111, 063061 (2025). arXiv:2410.03828 [gr-qc]

2025 **J. Feldbrugge**, "Phase-space Delaunay tessellation field estimator," *Monthly Notices of the Royal Astronomical Society* 536, 807–815 (2025). arXiv:2402.16234 [astro-ph.CO]

2023 **J. Feldbrugge**, Y. Yan, and R. van de Weygaert, "Statistics of tidal and deformation eigenvalue fields in the primordial Gaussian matter distribution: the two-dimensional case," *Monthly Notices of the Royal Astronomical Society* 526, 5031–5053 (2023). arXiv:2301.07200 [astro-ph.CO]

2023 **J. Feldbrugge**, "Complex evaluation of angular power spectra: Going beyond the Limber approximation," *Physical Review D* 108, 103007 (2023). arXiv:2304.13064 [astro-ph.CO]

2023 D. L. Jow, U.-L. Pen, and **J. Feldbrugge**, "Regimes in astrophysical lensing: refractive optics, diffractive optics, and the Fresnel scale," *Monthly Notices of the Royal Astronomical Society* 525, 2107–2124 (2023). arXiv:2204.12004 [astro-ph.HE]

2023 **J. Feldbrugge** and N. Turok, "Existence of real time quantum path integrals," *Annals of Physics* 454, 169315 (2023). arXiv:2207.12798 [hep-th]

2023 **J. Feldbrugge**, "Multiplane lensing in wave optics," *Monthly Notices of the Royal Astronomical Society* 520, 2995–3006 (2023). arXiv:2010.03089 [astro-ph.CO]

2023 **J. Feldbrugge**, U.-L. Pen, and N. Turok, "Oscillatory path integrals for radio astronomy," *Annals of Physics* 451, 169255 (2023). arXiv:1909.04632 [astro-ph.HE]

2023 **J. Feldbrugge** and R. van de Weygaert, "Cosmic web & caustic skeleton: non-linear constrained realizations – 2D case studies," *Journal of Cosmology and Astroparticle Physics* 2023(02), 058 (2023). arXiv:2212.07840 [astro-ph.CO]

2021 G. Wilding, K. Nevenzeel, R. van de Weygaert, G. Vegter, P. Pranav, B. J. T. Jones, K. Efstathiou, and **J. Feldbrugge**, "Persistent homology of the cosmic web – I. Hierarchical topology in ΛCDM cosmologies," *Monthly Notices of the Royal Astronomical Society* 507, 2968–2990 (2021). arXiv:2011.12851 [astro-ph.CO]

2019 A. Di Tucci, **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "Quantum incompleteness of inflation," *Physical Review D* 100, 063517 (2019). arXiv:1906.09007 [hep-th]

2019 **J. Feldbrugge**, M. van Engelen, R. van de Weygaert, P. Pranav, and G. Vegter, "Stochastic homology of Gaussian vs. non-Gaussian random fields: graphs towards Betti numbers and persistence diagrams," *Journal of Cosmology and Astroparticle Physics* 2019(09), 052 (2019). arXiv:1908.01619 [astro-ph.CO]

2019 P. Pranav, R. van de Weygaert, G. Vegter, B. J. T. Jones, R. J. Adler, **J. Feldbrugge**, C. Park, T. Buchert, and M. Kerber, "Topology and geometry of Gaussian random fields I: on Betti numbers, Euler characteristic, and Minkowski functionals," *Monthly Notices of the Royal Astronomical Society* 485, 4167–4208 (2019). arXiv:1812.07310 [astro-ph.CO]

2018 **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "Inconsistencies of the new no-boundary proposal," *Universe* 4, 100 (2018). arXiv:1805.01609 [hep-th]

2018 **J. Feldbrugge**, R. van de Weygaert, J. Hidding, and Joost Feldbrugge, "Caustic skeleton & cosmic web," *Journal of Cosmology and Astroparticle Physics* 2018(05), 027 (2018). arXiv:1703.09598 [astro-ph.CO]

2018 **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "No rescue for the no boundary proposal: Pointers to the future of quantum cosmology," *Physical Review D* 97, 023509 (2018). arXiv:1708.05104 [hep-th]

2017 **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "No smooth beginning for spacetime," *Physical Review Letters* 119, 171301 (2017). arXiv:1705.00192 [hep-th]

2017 **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "Lorentzian quantum cosmology," *Physical Review D* 95, 103508 (2017). arXiv:1703.02076 [hep-th]

2016 **J. Feldbrugge**, J. Hidding, and R. van de Weygaert, "Statistics of caustics in large-scale structure formation," *The Zeldovich Universe: Genesis and Growth of the Cosmic Web*, IAU Symposium 308, 107–114 (2016). arXiv:1412.5121 [astro-ph.CO]

2011 R. van de Weygaert, G. Vegter, H. Edelsbrunner, B. J. T. Jones, P. Pranav, C. Park, W. A. Hellwing, B. Eldering, N. Kruithof, E. G. P. Bos, J. Hidding, **J. Feldbrugge**, E. ten Have, M. van Engelen, M. Caroli, and M. Teillaud, "Alpha, Betti and the megaparsec universe: on the topology of the cosmic web," *Transactions on Computational Science XIV*, Lecture Notes in Computer Science 6970, 60–101 (2011). arXiv:1306.3640 [astro-ph.CO]

Preprints

2026 **J. Feldbrugge**, "Real-time path integral resurgence: which classical path contributes?" (2026). arXiv:2611.12510 [math-ph]

2026 S. Crew and **J. Feldbrugge**, "Asymptotic evaluation of multidimensional oscillatory integrals" (2026). arXiv:2610.22767 [math-ph]

2026 I. Aniceto, **J. Feldbrugge**, and C. J. Howls, "Which saddles contribute? The south-east rule for multidimensional integrals" (2026). arXiv:2606.28271 [math-ph]

2026 A. Read, **J. Feldbrugge**, C. Boehm, R. van de Weygaert, and B. Hertzsch, "Caustic skeleton and the local cosmic web: the Coma cluster node and the Pisces-Perseus ridge" (2026). arXiv:2604.22213 [astro-ph.CO]

2023 **J. Feldbrugge**, D. L. Jow, and U.-L. Pen, "Crossing singularities in the saddle point approximation" (2023). arXiv:2309.12427 [quant-ph]

2023 **J. Feldbrugge** and N. M. D. Niezink, "Orthogonality relations for conical functions of imaginary order" (2023). arXiv:2309.05616 [math.CA]

2020 **J. Feldbrugge** and N. Turok, "Gravitational lensing of binary systems in wave optics" (2020). arXiv:2008.01154 [gr-qc]

Theses

2019 **J. Feldbrugge**, "Path integrals in the sky: Classical and quantum problems with minimal assumptions," PhD thesis, Perimeter Institute, University of Waterloo, supervised by N. Turok. Available online.

2015 **J. Feldbrugge**, "Primordial non-Gaussianity and large-scale structure," Part III Essay, University of Cambridge, supervised by P. Shellard and T. Giannantonio. Available online.

2014 **J. Feldbrugge**, "Statistics of caustics in large-scale structure formation," Master thesis, University of Groningen, supervised by R. van de Weygaert, D. Roest, A.E. van Enter. Available online.

2012 **J. Feldbrugge** and M. van Engelen, "Analysis of Betti numbers and persistence diagrams of two-dimensional Gaussian random fields," Bachelor thesis, University of Groningen, supervised by R. van de Weygaert, E. Pallante, G. Vegter. Available online.

Invited talks

October 2026 Filaments in the multi-scale Universe: understanding their role and detectability, Kavli IPMU, Kashiwa, Japan
Presentation: Caustic walls and filaments in the cosmic web

August 2026 Astronomy Department of The Physics Institute of the Federal University of Rio Grande do Sul, Porto Alegre, Brazil
Seminar: Unravelling the cosmic web with caustics

July 2026 EXCOSM workshop: Random Structures and Patterns in Spatial Data: Here, There, and Everywhere, Nancy, France
Presentation: The caustic skeleton of the cosmic web

May 2026 School of Physics and Astronomy, University of Nottingham, Nottingham, United Kingdom
Seminar: Real-Time Path Integrals, Caustics and Interference in Cosmology

May 2026	The Center of Gravity – Niels Bohr Institute, Copenhagen, Denmark Seminar: Real-Time Path Integrals, Caustics and Interference in Cosmology
April 2026	Cosmological Frontiers in Fundamental Physics 2026, Perimeter Institute, Waterloo, Canada Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology
December 2025	King's College London, London, United Kingdom Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology
November 2025	University of Southampton, Southampton, United Kingdom Seminar: Unravelling the cosmic web with caustics
November 2025	University of Vienna, Vienna, Austria Presentation: Understanding the cosmic web with caustics
October 2025	Université Paris-Saclay, Paris, France Presentation: Understanding the cosmic web with caustics
October 2025	University of Bonn, Bonn, Germany Presentation: Real-Time Path Integrals, Caustics and Interference
October 2025	Université Paris-Saclay, Paris, France Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology
September 2025	Cosmology seminar at Perimeter Institute for Theoretical Physics, Waterloo, Canada Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology
August 2025	Seminar at the Canadian Institute for Theoretical Astrophysics, Toronto, Canada Presentation: Understanding the cosmic web with caustics
June 2025	The Cosmic Web from Galaxies to Cosmology, Institute for Fundamental Physics of the Universe, Trieste, Italy Presentation: Caustics and Interference
March 2025	Seminar at the University of Pennsylvania, Philadelphia, United States Presentation: Real-time Path Integrals, Caustics and Interference
October 2024	Complexity and Cosmos, Gran Sasso Science Institute, L'Aquila, Italy Presentation: Path integrals in the sky
June 2024	Singular and Oscillatory Integrals, University College London, London, United Kingdom Presentation: Integration in the complex plane with Picard-Lefschetz theory
May 2024	Tuorla-Tartu meeting 2024: Borderless Universe, Tuorla, Finland Presentation: What makes a cusp/filament?
December 2023	Gauge Gravity by the Ghats seminar, Center for High Energy Physics (CHEP), Indian Institute of Science, Bengaluru, India Presentation: Complex classical paths in quantum reflections and tunnelling
December 2023	Large-scale parity violation, Academia Sinica Institute of Astronomy and Astrophysics, Taipei, Taiwan Presentation: Dissecting the cosmic web with caustics
October 2023	International Loop Quantum Gravity Seminar (online) Presentation: Complex saddle points in gravitational path integrals
September 2023	Theoretical Physics seminar, Newcastle University, Newcastle, United Kingdom Presentation: On the existence of real-time path integrals
September 2023	Complexity and Cosmos, Gran Sasso Science Institute, L'Aquila, Italy Presentation: On the existence of real-time path integrals
July 2023	Quantum Gravity 2023, Radboud University, Nijmegen, The Netherlands Presentation: On the existence of real-time path integrals
July 2023	Theoretical physics group, University of New Brunswick, Fredericton, Canada Presentation: On the existence of real-time path integrals
March 2023	Optimal Transport Theory and Applications to Physics, École de Physique des Houches, France Presentation: Dissecting the cosmic web with caustics
September 2022	2nd Roman Juszkiewicz Symposium, Nicolaus Copernicus Astronomical Center, Warsaw, Poland Presentation: Dissecting the cosmic web with caustics
June 2022	Information Universe 4, University of Groningen, Groningen, The Netherlands Presentation: Dissecting the cosmic web with caustics
May 2022	Cosmology seminar, University of Oxford, Oxford, United Kingdom Presentation: Dissecting the cosmic web with caustics
May 2022	UK Cosmo Meeting 2022, Newcastle University, Newcastle, United Kingdom Presentation (keynote): Dissecting the cosmic web with caustics
October 2021	Higgs Hour, University of Edinburgh, Edinburgh, United Kingdom Presentation: Interference, caustics and oscillatory integrals
May 2021	Seminar, Universidad Nacional Autónoma de México, Mexico City, Mexico Presentation: Interference phenomena in lensing and quantum physics

May 2021 — Sirius A symposium 2021: To infinity and beyond, Groningen, The Netherlands
Presentation: The caustic skeleton of the cosmic web

October 2020 — Quantum & Gravity Seminar, Radboud University, Nijmegen, The Netherlands
Presentation: Lorentzian quantum cosmology

October 2020 — Pulsar group meeting, CITA, Toronto, Canada
Presentation: Multi-plane lensing and gravitational binary lensing in wave optics

November 2019 — Scintillometry 2019, Max Planck Institute for Radio Astronomy, Bonn, Germany
Presentation: Oscillatory path integrals for radio astronomy

September 2016 — CITA-PI day: Gravitational non-linear instability, CITA, Toronto, Canada
Presentation: Shocks in the early universe and gravitational waves

Contributed talks

June 2025 — Cosmology from Home 2025 (online)
Presentation: What makes a cosmic wall/filament in the caustic skeleton of the cosmic web?

June 2025 — Triangular Conference on Cosmological Frontiers in Fundamental Physics 2025, Paris, France
Poster: Real-Time Path Integrals

October 2024 — Scintillometry 2024, University of Central Florida, Orlando, United States
Presentation: Lensing by rotating stars

July 2024 — Relativistic effects and novel observables, University of Geneva, Geneva, Switzerland
Presentation: What makes a filament/wall?

July 2024 — Cosmology from Home 2024 (online)
Presentation: What makes a filament/wall?

June 2024 — Theoretical Modeling of the Large Scale Structure of the Universe, University of Edinburgh, Edinburgh, United Kingdom
Presentation: What makes a filament/wall?

March 2024 — Large Scale Structure group meeting, University of Cambridge, Cambridge, United Kingdom
Presentation: Caustic skeleton: what makes a filament in the cosmic web?

February 2023 — The Co-evolution of the Cosmic Web and Galaxies across Cosmic Time, Kavli Institute for Theoretical Physics, UC Santa Barbara, California, United States
Presentation: Dissecting the cosmic web with caustics

July 2022 — Cosmology from Home 2022 (online)
Presentation: Dissecting the cosmic web with caustics

March 2022 — Cosmic Cartography 2022: Exploring the Cosmic Web and Large-Scale Structure, Kavli IPMU, Kashiwa, Japan
Presentation: The caustic web and non-linear constrained Gaussian random fields

October 2021 — Tuorla-Tartu meeting 2021, University of Turku, Turku, Finland
Presentation: Caustic skeleton of the cosmic web

September 2020 — Cosmology from Home 2020 (online)
Presentation: The caustic skeleton of the cosmic web

March 2020 — Topological statistics group meeting, Department of Statistics, Carnegie Mellon University, Pittsburgh, United States
Two presentations: Cosmology and topology I, and Cosmology and topology II

February 2020 — Centre for the Universe and Waterloo Centre for Astrophysics day, Waterloo, Canada
Presentation: Path integrals for radio astronomy and gravitational lensing II

February 2020 — Cosmology group meeting, Perimeter Institute, Waterloo, Canada
Presentation: The caustic skeleton of the cosmic web

January 2020 — The cosmic web in the local universe, Lorentz Center, Leiden, The Netherlands
Presentation: The caustic skeleton of the cosmic web

October 2019 — The Future of Astronomy, Waterloo Centre for Astrophysics, Waterloo, Canada
Poster: Oscillatory path integrals for radio astronomy

September 2019 — Theory group meeting, Carnegie Mellon University, Pittsburgh, United States
Presentation: Interference and Picard-Lefschetz theory

September 2019 — Simplicity III, Perimeter Institute, Waterloo, Canada
Presentation: Fun with path integrals II

August 2019 — Graduate student meeting, Perimeter Institute, Waterloo, Canada
Presentation: Oscillatory integrals in the complex plane

June 2019 — Probabilities in cosmology, University of Groningen, The Netherlands
Presentation: Lorentzian beginnings of the universe

June 2019	The cosmic web: from galaxies to cosmology, Edinburgh, United Kingdom Presentation: Caustic skeleton of the cosmic web
May 2019	Cosmology group meeting Perimeter Institute, Waterloo, Canada Presentation: Lenses and oscillatory integrals
May 2018	Albert Einstein Institute group meeting, Potsdam, Germany Presentation: Classical and weak trajectories
November 2017	Path integral of gravity, Perimeter Institute, Waterloo, Canada Presentation: Quantum incompleteness of inflation II
September 2017	Cosmology group meeting CITA, Toronto, Canada Presentation: The instability of the no-boundary proposal
July 2017	Cosmic web day, University of Toronto, Toronto, Canada Presentation: The caustic skeleton of the cosmic web
May 2017	PI-day, Perimeter Institute, Waterloo, Canada Presentation: Lorentzian quantum cosmology
May 2017	Theory Canada 12, York University, Toronto, Canada Presentation: Lorentzian quantum cosmology
May 2017	Cosmology group meeting Perimeter Institute, Waterloo, Canada Presentation: Caustics in large-scale structure
May 2017	String cosmology group meeting Van Swinderen Institute, Groningen, The Netherlands Presentation: Lorentzian quantum cosmology
April 2017	New Thoughts 3: About the universe and more, Ely, United Kingdom Presentation: Lorentzian quantum cosmology
April 2017	British gravity meeting 2017, University of Oxford, Oxford, United Kingdom Presentation: Lorentzian quantum cosmology
June 2016	Cosmology group meeting CITA, Toronto, Canada Presentation: Statistics of caustics in large-scale structure
May 2016	Cosmology group meeting Perimeter Institute, Waterloo, Canada Presentation: Statistics of caustics in large-scale structure
March 2016	Statistics of extrema of large-scale structure, Lorentz Center, Leiden, The Netherlands Presentation: Statistics of caustics in large-scale structure
June 2014	IAUS 308: The Zel'dovich universe, Tallinn, Estonia Presentation: Statistics of caustics in large-scale structure formation
October 2012	Structure of the cosmic web, Leibniz Institute for Astrophysics, Potsdam, Germany Presentation: Analysis of Betti numbers and persistence diagrams in 2D GRFs

Conferences and workshops (attended only)

March 2026	Max Planck-IAS-NTU Center Kick-off Conference, Institute for Advanced Study, Princeton, United States
August 2025	Charting the Future Symposium: Big questions in particle physics, strong gravity, and cosmology over the next 25 years, Perimeter Institute, Waterloo, Canada
June 2022	Analogue Models of Gravity and Fluctuation-Induced phenomena, University of Edinburgh, Edinburgh, United Kingdom
June 2022	Online workshop: Physics of the Early Universe
January 2021	Cosmology 2021: the rise of field theory, University of Cambridge, Cambridge, United Kingdom
October 2020	The information universe: What is the role of information in our Universe? University of Groningen, Groningen, The Netherlands
January 2020	First Dutch Mathematical Relativity Day, Radboud University Nijmegen, Nijmegen, The Netherlands
September 2019	Cosmological frontiers in fundamental physics 2019, Perimeter Institute, Waterloo, Canada
November 2018	Quantum universe, in celebration of Neil Turok's 60th birthday, Centro de Estudios Científicos (CECs), Valdivia, Chile
August 2018	Cosmology and gravitational physics with lambda, Nordita, Stockholm, Sweden
June 2018	Scanning new horizons: Emergent space-time, black holes and quantum information, Van Swinderen Institute, Groningen, The Netherlands
January 2018	Gravity in the early universe, Princeton University, Princeton, United States
June 2017	Bounce scenarios in cosmology, Perimeter Institute, Waterloo, Canada
January 2017	Fundamentals of the universe, Van Swinderen Institute, Groningen, The Netherlands
January 2017	New directions in theoretical physics II, Higgs Centre for Theoretical Physics, Edinburgh, United Kingdom
October 2016	Midwest relativity meeting, Perimeter Institute, Waterloo, Canada
June 2016	Time in cosmology, Perimeter Institute, Waterloo, Canada
June 2016	Concepts and paradoxes in a quantum universe, Perimeter Institute, Waterloo, Canada

June 2016 Cosmological frontiers in fundamental physics 2016, Perimeter Institute, Waterloo, Canada
June 2015 Convergence, Perimeter Institute, Waterloo, Canada
April 2014 Quantum universe 4, University of Groningen, The Netherlands
March 2013 Quantum universe 3, University of Groningen, The Netherlands

Software

Open-source codes that accompany my research.
IntegralResurgence.jl — Asymptotic evaluation of multidimensional oscillatory integrals using resurgence: Stokes constants, the adjacency of saddle points, and superasymptotic and hyperasymptotic approximations. Accompanies arXiv:2610.22767. Registered in the Julia General registry. GitHub · Documentation · Zenodo
PicardLefschetzIntegration.jl — Numerical evaluation of multidimensional oscillatory integrals using Picard–Lefschetz theory, with applications to wave optics in astronomy. Registered in the Julia General registry. GitHub · Documentation · Zenodo
Picard–Lefschetz Path Integrals (C++) — Evaluation of multidimensional oscillatory integrals using Picard–Lefschetz theory, demonstrated with lensing problems in wave optics; with U.-L. Pen and N. Turok. Accompanies Annals of Physics 451, 169255 (2023). Website
PhaseSpaceDTFE.jl — Density and velocity fields of N-body simulations with the phase-space Delaunay tessellation field estimator. Accompanies MNRAS 536, 807 (2025). Registered in the Julia General registry. GitHub · Documentation · Zenodo
Nbody2D.jl — Particle-mesh code for two-dimensional cosmological N-body simulations with unconstrained and constrained Gaussian random field initial conditions. Registered in the Julia General registry. GitHub · Documentation · Zenodo
DiscreteNaturalNeighbors.jl — Discrete natural neighbour interpolation of n-dimensional data, developed by Benjamin Hertzsch in collaboration with me. Registered in the Julia General registry. GitHub · Documentation
HHGsimPL.jl — Simulation of high-order harmonic generation using Picard–Lefschetz integration methods, developed by Anne Weber in collaboration with me. Accompanies Physical Review A 113, 063111 (2026). GitHub · Zenodo

Advising

PhD projects

2025 - present Advisor PhD project, Maé Rodriguez, University of Edinburgh, Edinburgh, United Kingdom: **Evaluating real-time path integrals for relativistic systems**
2025 - present Advisor PhD project, Amelie Read, University of Sydney, Sydney, Australia: **The caustic skeleton of the local cosmic web**
2024 - 2025 Co-advisor PhD project, Leigh Preimesberger, Carnegie Mellon University, Pittsburgh, United States: **Machine learning inference of the cosmic web**
2023 - present Advisor PhD project, Benjamin Hertzsch, University of Edinburgh, Edinburgh, United Kingdom: **Walls, filaments and galaxy populations in the caustic skeleton**
2023 - 2026 Co-advisor PhD project, Anne Weber, University College London, London, United Kingdom: **Complex trajectories in laser ionization and recombination experiments**
2023 - 2025 Co-advisor PhD project, Johanna Borissova, Perimeter Institute, Waterloo, Canada: **Lorentzian wormholes**
2023 - 2025 Co-advisor PhD project, Ariadna Metidieri, Radboud University, Nijmegen, The Netherlands: **Lensing of rotating stars and photons in wave optics**
2022 - present Co-advisor PhD project, Joshua Jones, Dublin Institute for Advanced Studies, Dublin, Ireland: **Real-time path integrals**
2020 - 2022 Co-advisor PhD project, Georg Wilding, University of Groningen, The Netherlands: **Topology of the two-dimensional cosmic web**
2020 - 2021 Co-advisor PhD project, Varun Rustagi, University of Groningen, The Netherlands: **The caustic skeleton of the local universe**

Master projects

2026 - present Collaboration with Joanna Chan, independent scientist, Singapore: **Properties of the caustic skeleton in scale space**

2026 - present Collaboration with Guilherme Garcia, master student at the University of Copenhagen, Copenhagen, Denmark

2025 - 2026 Advisor master project, Guilherme Garcia, University of Edinburgh, Edinburgh, United Kingdom: **Constrained non-Gaussian fields**

2025 - 2026 Advisor master project, Joanna Chan, University of Edinburgh, Edinburgh, United Kingdom

2024 Advisor master project, Maé Rodriguez, University of Edinburgh, Edinburgh, United Kingdom: **Hamiltonian Monte Carlo sampling of primordial caustic constraints**

2023 Advisor master project, Vera Li, University of Edinburgh, Edinburgh, United Kingdom: **The caustic skeleton and redshift space distortions**

2022 - 2023 Advisor master project, Yihan Yan, University of Waterloo, Canada: **Statistics of critical points in eigenvalue fields**

2018 - 2022 Co-advisor master project, Kevin Bixerman, University of Groningen, The Netherlands: **The caustic skeleton of the local universe**

Bachelor projects

2024 - 2025 Advisor bachelor project (Senior Honours), Guilherme Garcia, University of Edinburgh, Edinburgh, United Kingdom: **The cosmic web**

2022 - 2023 Co-advisor bachelor project, Yonatan Sklansky, University of Pennsylvania, United States: **Topology of multi-stream regions in N-body simulations**

2021 - 2022 Advisor bachelor project, Yihan Yan, University of Waterloo, Canada: **Homology of two- and three-dimensional Gaussian random fields**

Teaching

July 2025 Lecturer, EXCOSM Summer School, Large-scale structure of the Universe: from galaxies to cosmology, Haapsalu, Estonia

June 2023 Lecturer, School on Fundamentals of the Universe, Lorentz Center, Leiden, The Netherlands

June 2018 2018 Rotman summer institute in philosophy of cosmology, Goderich, Canada
Two lectures on Lorentzian quantum cosmology

January 2018 PSI winter school, co-supervisor project 'Pair creation in de Sitter spacetime', Huntsville, Canada

2010 - 2015 Physics lecturer, exam training for secondary school students, UOCG Market BV

2013 - 2014 Developer of teaching material for exam training in physics, UOCG Market BV

2008 - 2014 Tutoring in mathematics and physics for secondary school students

Service

2022 - 2023 Co-organizer of the conference New Directions in Theoretical Physics 4 (January 2023), University of Edinburgh, Edinburgh, United Kingdom

2018 - present Reviewer for Physical Review Letters (PRL), Physical Review D (PRD), the Journal of Cosmology and Astroparticle Physics (JCAP), the Journal of High Energy Physics (JHEP), Monthly Notices of the Royal Astronomical Society (MNRAS), and Universe.

2013 - 2014 Chairman of the Professor Hendrik de Waard Foundation

2012 - 2013 Treasurer of the Professor Hendrik de Waard Foundation

2011 - 2012 Member of the education committee mathematics, chairman of the student council

2010 - 2014 Guide at the Gratama telescope of the Blaauw observatory, Groningen, at stargazing events

Languages

Dutch	Mother tongue	English	Fluent
German	Elementary	French	Elementary
Latin	Elementary		

Programming experience

Mathematica	Fluent	C++	Fluent
Python	Fluent	Matlab	Moderate
Swift	Moderate	Julia	Fluent
Fortran	Moderate	R	Elementary